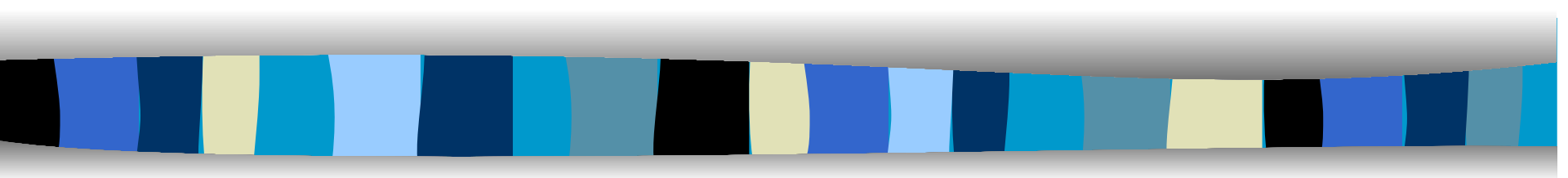


Asymmetric Cryptography

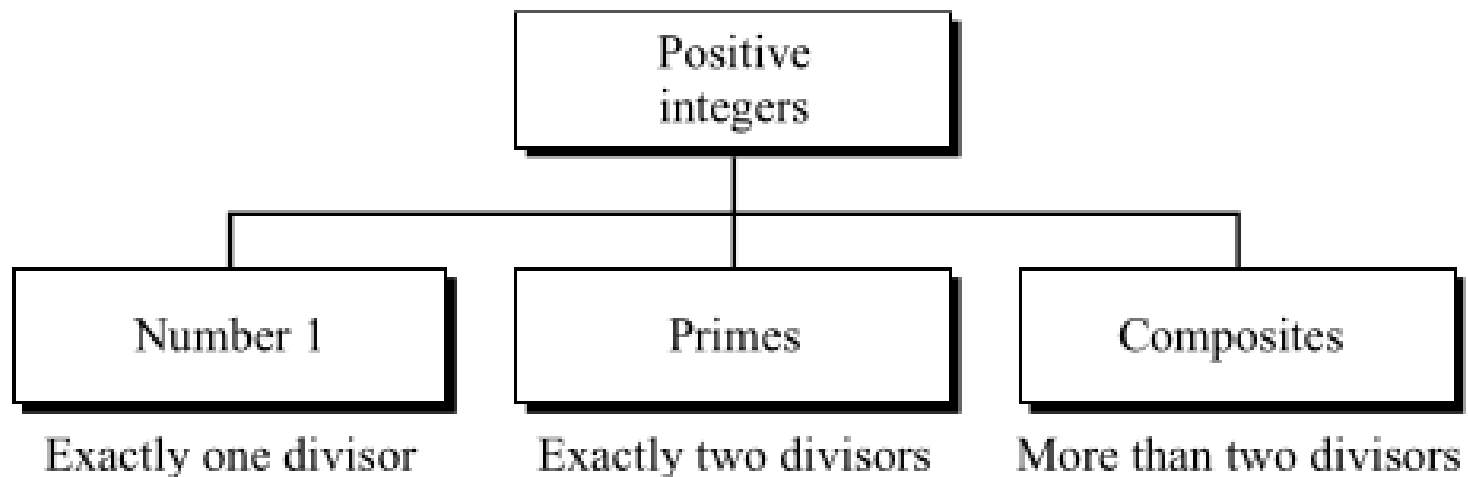


Number Theory

Primes

- Asymmetric-key cryptography uses primes extensively.
- A prime number is divisible by itself or by 1.

Three groups of positive integers



Co-primes

- Two positive integers, a and b , are relatively prime, or coprime, if $\gcd(a, b) = 1$.
- Note that the number 1 is relatively prime with any integer.
- If p is a prime, then all integers 1 to $p - 1$ are relatively prime to p .
- Z_{n^*} is the set of integers less than n which are all relatively prime to n .
- Set Z_{p^*} is the same except that modulus (p) is a prime.

Cardinality of prime

- There are infinite number of primes.
- So, cardinality of primes is infinite.

Checking primeness

- Check if n is prime
- $q = \text{floor}(\text{sqrt}(n))$
- Check if any primes less than q divides n
- If so, then n is not a prime
- Else, n is prime.
- Check if 97 is prime
- $q = \text{floor}(\text{sqrt}(97)) = 9$
- The primes less than 9 are 2, 3, 5, 7
- None of these divides 97. So, 97 is prime.

Euler's phi function

- Euler's phi-function, $\phi(n)$, also known as the Euler's totient function.
- It has a very important role in cryptography.
- The function finds the number of integers that are both smaller than n and relatively prime to n .
- So, it helps to find cardinality of set Z_{n^*}
- The function $\phi(n)$ is defined as

Euler's phi function

- $\varphi(1) = 0$.
- $\varphi(p) = p - 1$ if p is a prime.
- $\varphi(m \times n) = \varphi(m) \times \varphi(n)$ if m and n are relatively prime.
- $\varphi(p^a) = p^a - p^{a-1}$ if p is a prime and a is some integer
- $\varphi(p_1^a * p_2^b) = (p_1^a - p_1^{a-1}) * (p_2^b - p_2^{b-1})$

Euler's phi function

- $\varphi(13)=13-1=12$ $Z_{13^*}=\{1,2,3,4,5,6,7,8,9,10,11,12\}$
- $\varphi(10)=\varphi(2*5)=\varphi(2)*\varphi(5)=(2-1)*(5-1)=4$
- $Z_{10^*}=\{1,3,7,9\}$
- $\varphi(240)=\varphi(24*10)=\varphi(2^3*3*2*5)=\varphi(2^4*3*5)$
- $= (2^4-2^{4-1})*(3^1-3^{1-1})*(5^1-5^{1-1})=(16-8)*2*4=64$

Format's little theorem

- There are two versions of Format's little theorem.
- The first version says that if p is a prime and a is a positive integer such that p does not divide a , then
- $a^{p-1} \equiv 1 \pmod{p}$.
- This is true for all positive integers $a \neq p$

Format's little theorem

- Let $p=5$
- $1^4 \bmod 5=1$
- $2^4 \bmod 5=1$
- $3^4 \bmod 5=1$
- $4^4 \bmod 5=1$
- $6^4 \bmod 5=1$
- $7^4 \bmod 5=1$
- $8^4 \bmod 5=1$

Format's little theorem

- Let $p=7$
- $1^6 \bmod 7=1$
- $2^6 \bmod 7=1$
- $3^6 \bmod 7=1$
- $4^6 \bmod 7=1$
- $6^6 \bmod 7=1$
- $8^6 \bmod 7=1$
- $9^6 \bmod 7=1$

Format's little theorem

- Another version is
- $a^p \equiv a \pmod{p}$.
- Where p is prime, a is positive integer $a < p$

Format's little theorem

- Let $p=5$
- $1^5 \bmod 5=1$
- $2^5 \bmod 5=2$
- $3^5 \bmod 5=3$
- $4^5 \bmod 5=4$

Format's little theorem

- Let $p=7$
- $1^7 \bmod 7=1$
- $2^7 \bmod 7=2$
- $3^7 \bmod 7=3$
- $4^7 \bmod 7=4$
- $5^7 \bmod 7=5$
- $6^7 \bmod 7=6$

Format's little theorem

- $6^{10} \bmod 11 = 1$
- Find $3^{12} \bmod 11$
- $= 3^{11} * 3 \bmod 11$
- $= [(3^{11} \bmod 11) * (3 \bmod 11)] \bmod 11$
- $= [3 * 3] \bmod 11$
- $= 9$

Inverse using Fermat's little theorem

- We can find multiplicative inverse using Fermat's little theorem.
- $a \times a^{-1} \bmod p$
- $= a \times a^{-1} \times 1 \bmod p$
- $= a \times a^{-1} \times a^{p-1} \bmod p$
- $= a \times a^{p-2} \bmod p$
- $= a^{p-1} \bmod p$
- $= 1$

Inverse using Format's little theorem

- Find the 8^{-1} modulo 17.
- $8^{-1} \bmod 17$
- $=8^{17-2} \bmod 17$
- $=8^{15} \bmod 17$, $a=8$, $m=15$, $n=17$, $z=1$
- Here, 15 in binary is 1111

i	B_i of m	$z=z^2 \bmod n$	$z=z*a \bmod n$
3	1	1	$=1*8 \bmod 17=8$
2	1	$8^2 \bmod 17=13$	$=13*8 \bmod 17=2$
1	1	$2^2 \bmod 17 = 4$	$=4 *8 \bmod 17 =15$
0	1	$15^2 \bmod 17 =4$	$=4 *8 \bmod 17 =15$

Inverse using Fermat's little theorem

- Find the 5^{-1} modulo 23.
- 5^{-1} modulo 23.
- $=5^{23-2} \bmod 23$
- $=5^{21} \bmod 23$
- 21 in binary is 10101 which is 16, 4, 1
- $5^{21} \bmod 23 = 5^{16} \times 5^4 \times 5 \bmod 23$
- $5^2 \bmod 23 = 2$
- $5^4 \bmod 23 = (5^2)^2 \bmod 23 = (5^2 \bmod 23)^2 \bmod 23$
- $= 2^2 \bmod 23 = 4$

Inverse using Fermat's little theorem

- Find the 5^{-1} modulo 23.
- $5^8 \bmod 23 = (5^4)^2 \bmod 23 = (5^4 \bmod 23)^2 \bmod 23$
- $= 4^2 \bmod 23 = 16$
- $5^{16} \bmod 23 = (5^8)^2 \bmod 23 = (5^8 \bmod 23)^2 \bmod 23 = 16^2 \bmod 23 = 3$
- Now, $5^{21} \bmod 23 = 5^{16} \times 5^4 \times 5 \bmod 23$
- $= 3 \times 4 \times 5 \bmod 23 = 60 \bmod 23 = 14$

Euler's Theorem

- Euler's theorem can be thought of as a generalization of Fermat's little theorem.
- The modulus in the Fermat theorem is a prime, the modulus in Euler's theorem is an integer.
- We introduce two versions of this theorem.
- The first version is
- If a and n are coprime, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.
- It means, If power of a is $\varphi(n)$ then its remainder is 1.
- Where $\varphi(n)$ is totient function.

Euler's Theorem

- Find the result of $6^{24} \bmod 35$.
- Here 6 and 35 are co-prime.
- So, $6^{24} \bmod 35 = 6^{\phi(35)} \bmod 35$
- $\phi(35) = \phi(7 \cdot 5) = 6 \cdot 4 = 24$
- 1

Euler's Theorem

- The second version of Euler's theorem removes the condition that a and n should be coprime.
- $a < n$, then
- $a^{\varphi(n) + 1} \equiv a \pmod{n}$.
- Find the result of $20^{62} \bmod 77$.
- $\varphi(77) = \varphi(11 \cdot 7) = 10 \cdot 6 = 60$
- $20^{62} \bmod 77 = 20^{\varphi(77)+1} * 20 \bmod 77$
- $[(20^{\varphi(77)+1} \bmod 77) * (20 \bmod 77)] \bmod 77$
- $[20 * 20] \bmod 77$
- $= 15$

Euler's Theorem

- The second version of Euler's theorem removes the condition that a and n should be coprime.
- If $n = p \times q$, $a < n$, and k an integer, then
- $a^{\varphi(n) + 1} \equiv a \pmod{n}$.
- Find the result of $20^{62} \bmod 77$.
- $\varphi(77) = \varphi(11 \times 7) = 10 \times 6 = 60$
- $20^{62} \bmod 77 = 20^{\varphi(77)+1} * 20 \bmod 77$
- $[(20^{\varphi(77)+1} \bmod 77) * (20 \bmod 77)] \bmod 77$
- $[20 * 20] \bmod 77$
- $= 15$

Euler's Theorem

- We can use Euler's theorem to calculate multiplicative inverse.
- $a \times a^{-1} \bmod n$
- $= a \times a^{-1} \times 1 \bmod n$
- $= [(a \times a^{-1} \bmod n) (1 \bmod n)] \bmod n$
- $= [(a \times a^{-1} \bmod n) (a^{\varphi(n)} \bmod n)] \bmod n$
- $= a \times a^{-1} \times a^{\varphi(n)} \bmod n$
- $a \times a^{\varphi(n) - 1} \bmod n$
- $= a^{\varphi(n)} \bmod n$
- $= 1$

Euler's Theorem

- Find $60^{-1} \bmod 187$
- $60^{-1} \bmod 187$
- $=60^{\varphi(187)-1} \bmod 187$
- $\varphi(187)=\varphi(11*17)=10*16=160$
- Now let come back
- $60^{\varphi(187)-1} \bmod 187$
- $=60^{159} \bmod 187$
- $a=60, m=159, n=187, z=1$

Euler's Theorem

- $a=60$, $m=159$, $n=187$, $z=1$

i	b_i of m	$z=z^2 \bmod n$	$z=z*a \bmod n$
7	1	$1^2 \bmod 187=1$	$1*60 \bmod 187=60$
6	0	$60^2 \bmod 187=47$	$=47$
5	0	$47^2 \bmod 187=152$	$=152$
4	1	$152^2 \bmod 187=103$	$=103*60 \bmod 187=9$
3	1	$9^2 \bmod 187=81$	$=81*60 \bmod 187=185$
2	1	$185^2 \bmod 187=4$	$=4*60 \bmod 187=53$
1	1	$53^2 \bmod 187=4$	$=4*60 \bmod 187=53$
0	1	$53^2 \bmod 187=4$	$=4*60 \bmod 187=53$

Miller Robin's Primality Test

- Miller Robin's primality test is based on fermat's little theorem.
- For a prime p and any positive integer a following is true
- $a^{p-1} \equiv 1 \pmod{p}$.
- Here p does not divide a .
- It generates a positive integer a such that $1 < a < p-1$
- Test p using miller robin's primality test with chosen a .
- Miller robin's test can return sure composite or probable prime

Miller Robin's Primality Test

- If it returns probable prime, then generate a for sufficient number of time s and test p for primality again.
- If it returns probable prime for s times then we can say p is prime with probability $1-1/4^s$
- For $s=10$, probability > 0.9999

Miller Robin's Primality Test

- Test_prime(){
- INPUT n
- IF $n \leq 1$, return “neither prime nor composite”
- ELSE IF $n < 3$ return “Prime”
- ELSE IF n is even, return “Composite”
- Choose sufficiently large s
- FOR i= 1 to s
- Generate random integer $1 < a < n-1$
- If miller_rabin_test(n,a)=“composite”, return “composite”
- END FOR
- RETURN “prime”}

Miller Robin's Primality Test

- miller_rabin_Test(n, a) {
- Find an positive integer k and positive odd integer m such that $n-1 = m \times 2^k$
- $t = a^m \bmod n$
- IF $t=1$ or $t=n-1$, return “probable prime”
- FOR $i = 1$ to $k-1$
- $t = t^2 \bmod n$
- If $t=n-1$ return “probable prime”
- END FOR
- RETURN “composite” }

Miller Robin's Example

- Lets test $n=561$
- $561-1=m \times 2^k$
- Or, $m=560/2^k$
- Put $k=1$, $m=560/2=280$
- Put $k=2$, $m=560/4=140$
- Put $k=3$, $m=560/8=70$
- Put $k=4$, $m=560/16=35$
- Put $k=5$, $m=560/32=17.5$
- So, $k=4$ and $m=35$

Miller Robin's Example

- Let randomly generated integer be $a=2$
- $T = 2^{35} \bmod 561 = 263$

i	t
1	$263^2 \bmod 561 = 166$
2	$166^2 \bmod 561 = 67$
3	$67^2 \bmod 561 = 1$

- So, 561 is composite

Miller Robin's Example

- Lets test $n=29$
- $29-1=7 \times 2^2$, which means $m=7$ and $k=2$
- Let randomly generated integer be $a=10$
- $t = 10^7 \bmod 29 = 10^2 \cdot 10^2 \cdot 10^2 \cdot 10 \bmod 29 = 17$

i	$t = t^2 \bmod n$
1	$17^2 \bmod 29 = 28$

- So, 29 is probable prime

Miller Robin's Example

- Lets test $n=29$
- $29-1=7 \times 2^2$, which means $m=7$ and $k=2$
- Let randomly generated integer be $a=3$
- $t = 3^7 \bmod 29 = 3^4 \cdot 3^3 \bmod 29 = 23 \cdot 27 \bmod 29 = 12$

i	$t = t^2 \bmod n$
1	$12^2 \bmod 29 = 28$

- So, 29 is still probable prime

Primitive root

- A number a is a primitive root modulo n if every number co-prime to n is congruent to power of a modulo n .
- The a is said to be primitive root of n if $a \bmod n$, $a^2 \bmod n$, $a^3 \bmod n$,, $a^{n-1} \bmod n$ are distinct.
- Is 2 a primitive root of 5?
- $2^1 \bmod 5 = 2$
- $2^2 \bmod 5 = 4$
- $2^3 \bmod 5 = 3$
- $2^4 \bmod 5 = 1$
- So, 2 is primitive root of 5

Logarithm

- It is always easy to compute exponent of a number powered with some number.
- $2^5=32$, $2^{100}=1267650600228229401496703205376$
- How to compute the exponent given its base and result.
- For example, $2^x=32$, here $x=5$
- $3^x=129140163$
- Computing exponent satisfying some relation is difficult for large values.
- We can use logarithm in such case.
- $x=\log_3(129140163)=17$

Discrete Logarithm

- In similar way, we can use logarithm to find the exponent in modular arithmetic.
- In such case, logarithm is called discrete logarithm since it always provides discrete values.
- It is represented with $\text{dlog}_{a,p}$
- Here, a is base and p is modulus.
- For example
- $b = a^i \bmod p$
- $i = \text{dlog}_{a,p}(b) = \log_a b \bmod p$
- Computing discrete logarithm is difficult as we need to try all the powers.

Discrete logarithm

- So, we can use this property for cryptography.
- It will be more difficult if discrete logarithm provides unique value.
- This is only possible if the base a is primitive root of p .
- Every discrete logarithm may not have solutions.
- $x = \log_3 4 \bmod 13$ has no solution as 3 is not primitive root of 13.
- The equation $b = a^i \bmod p$ has only one solution if b is relative prime to p and a is primitive root to p .
- $i = \text{dlog}_{a,p}(b)$

Discrete logarithm

- Find x in each of the following cases:
 - a. $4 \equiv 3^x \pmod{7}$.
 - b. $6 \equiv 5^x \pmod{7}$.
- Solution
- We can easily use the tabulation of the discrete logarithm in Table 9.6.
- a. $4 \equiv 3^x \pmod{7} \rightarrow x = L_3 4 \pmod{7} = 4 \pmod{7}$
- b. $6 \equiv 5^x \pmod{7} \rightarrow x = L_5 6 \pmod{7} = 3 \pmod{7}$

y	1	2	3	4	5	6
$x = L_3 y$	6	2	1	4	5	3
$x = L_5 y$	6	4	5	2	1	3